

ABR Auditory Stimulus Project

Kingshuk Daschowdhury, Eins Besmanos, Achyut Chebiyam, Jack Langhoff

The Biomedical Motivation: Preventing Hearing Loss

- 1 to 3 babies out of 1000 babies are born with permanent hearing loss.
- An infant's social development gets stunted at an early age due to this
- *Imagine not being able to hear your parents' voices growing up! Pretty sad.*

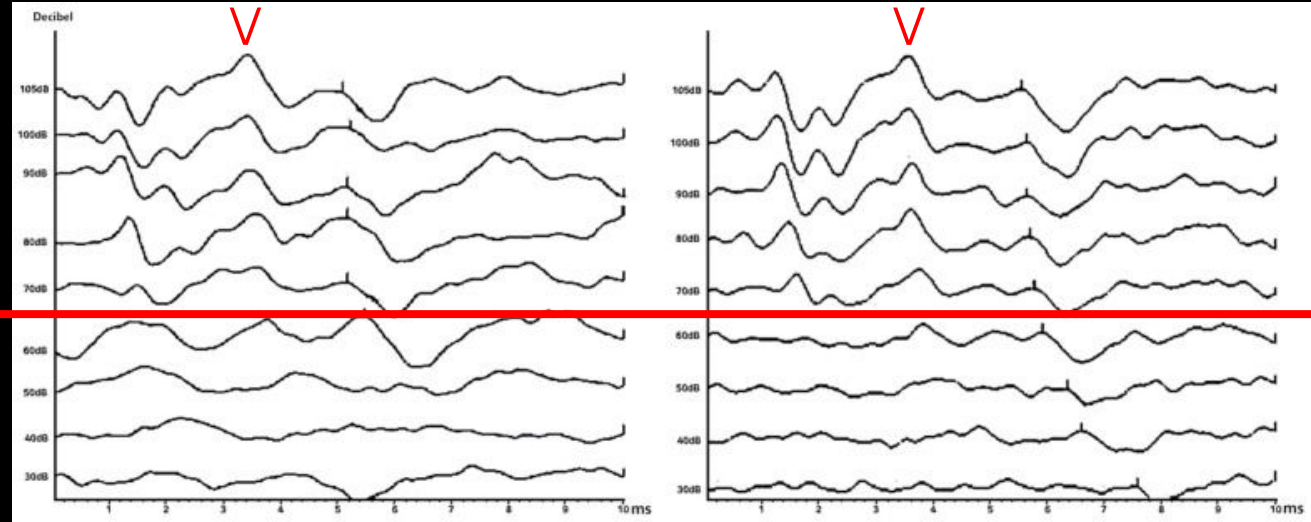


What does an ABR device do for early signs of hearing loss?

- Auditory Brainstem Response (ABR) is the **GOLD** standard for diagnosis
- ABR detects activity from the brain in response to sound stimuli.
- Absence of V waves confirms the newborn with congenital hearing loss.
- Intervention via hearing aids or cochlear implants

Patient w/ hearing loss (Left) vs Patient w/o hearing loss (right)

Starting under the red line (>70 dB), the V wave begins to dissipate on the left graph



Engineering Design Goals

Physiologic Need

- Detect V Wave
- Biosignal of $0.1\text{--}1\ \mu\text{V}$
- 60 Hz EMI
- Wave V Latency 5–7 ms
- Quantization $<$ Noise floor
- Output SNR

Engineering Specs

- BW: 100 Hz – 3 kHz
- Gain = 10^5 (100 dB)
- CMRR $>$ 100 dB
- Sampling rate $f_s = 20\text{ kHz}$ ($\gg 2 \cdot \text{BW}$)
- ADC: 12 bits (LSB = $2.4\text{ mV} \ll 280\text{ mV } \sigma_{\text{out}}$)
- Target SNR $>$ 20 dB after averaging



Takeaway: humans are unpredictable, especially babies!

Electrodes

Bandpass Filters

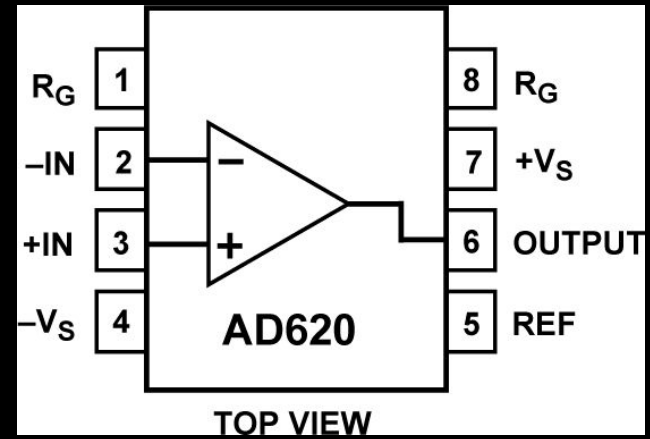
AAF

ADC

Computer

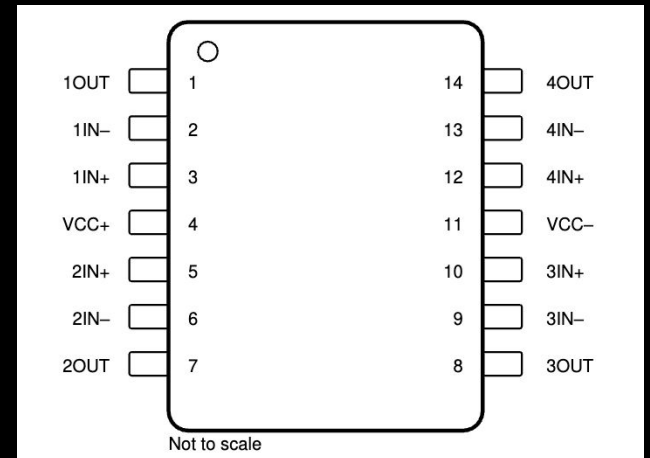
AD620 Sensor specs

- BW: 120 kHz
- Input Voltage Noise: 9 nV/ $\sqrt{\text{Hz}}$ @ 1 kHz
- Input Current Noise: 0.1 pA/ $\sqrt{\text{Hz}}$
- CMRR: 110 dB to 130 dB ($G > 1000$)
- \$13 to \$18

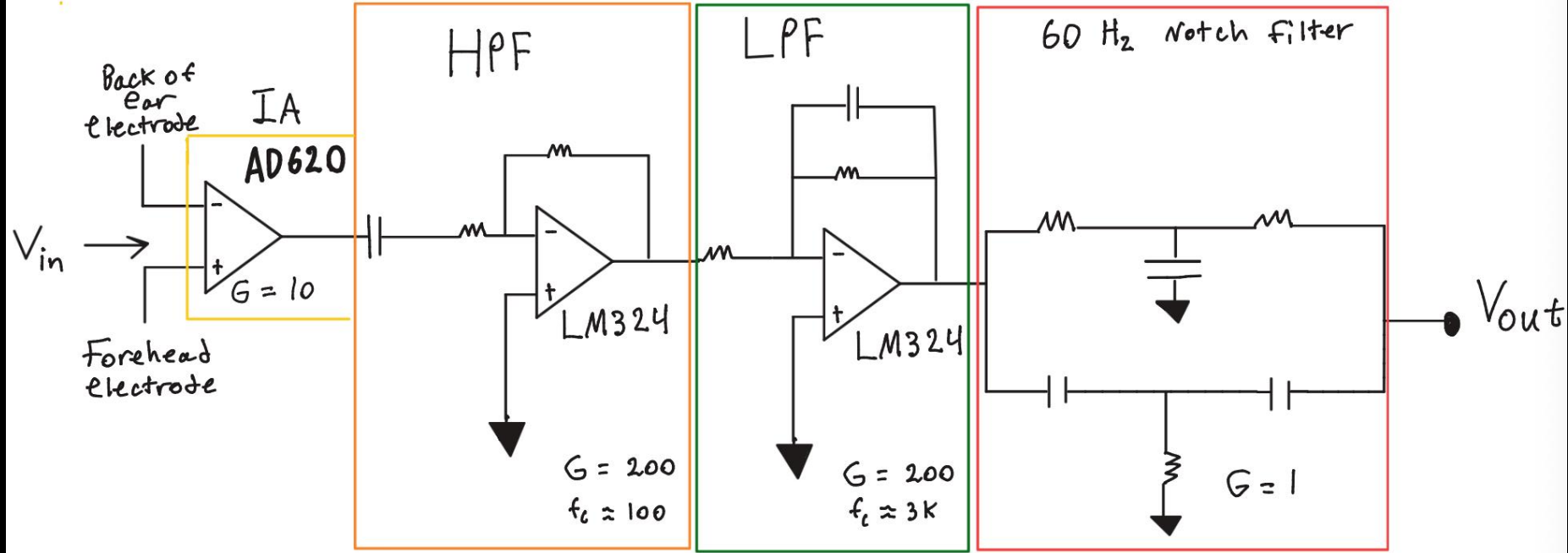


LM324 Sensor specs

- BW: 1 MHz
- Input Voltage Noise: 35 nV/ $\sqrt{\text{Hz}}$ @ 1 kHz
- Input Current Noise: 0.01 pA/ $\sqrt{\text{Hz}}$
- CMRR: 65 dB to 80 dB
- \$0.13 to \$0.60



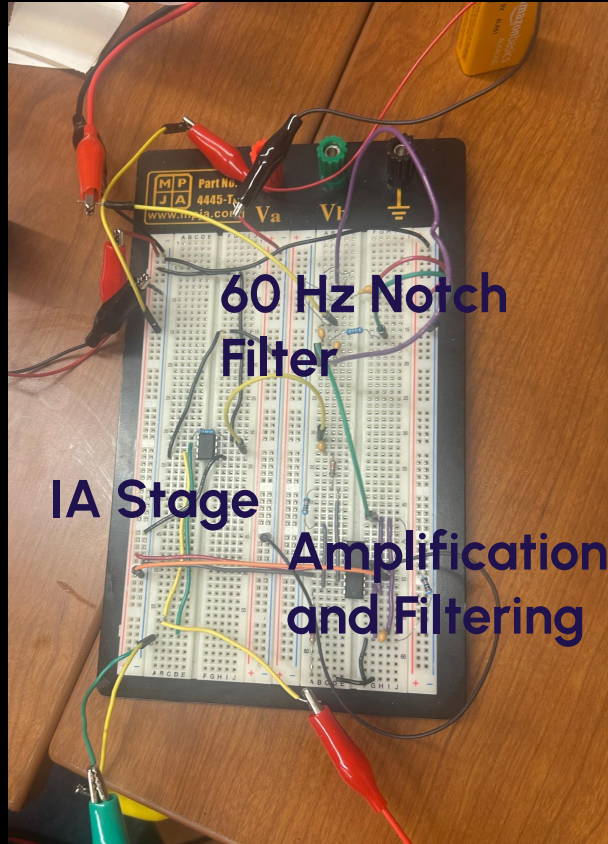
Circuit Diagram and Component Values



Total Gain = $10 \times 200 \times 200 \times 1 = 400000$

Cutoff Frequencies = 100 Hz - 2950 Hz

Data Acquisition Setup and Picture of Circuit



Sensor System and Noise and Interference Model

AD620 Source impedance = 90 kOhms

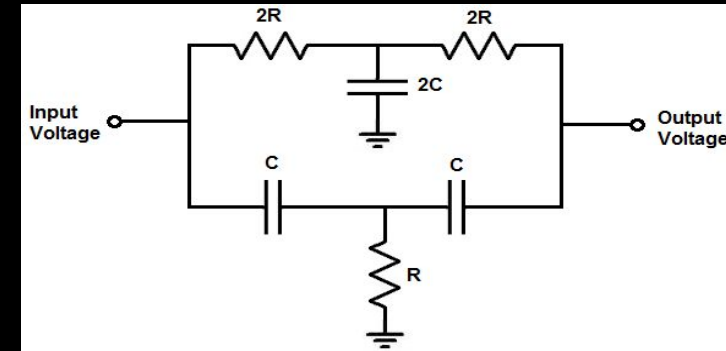
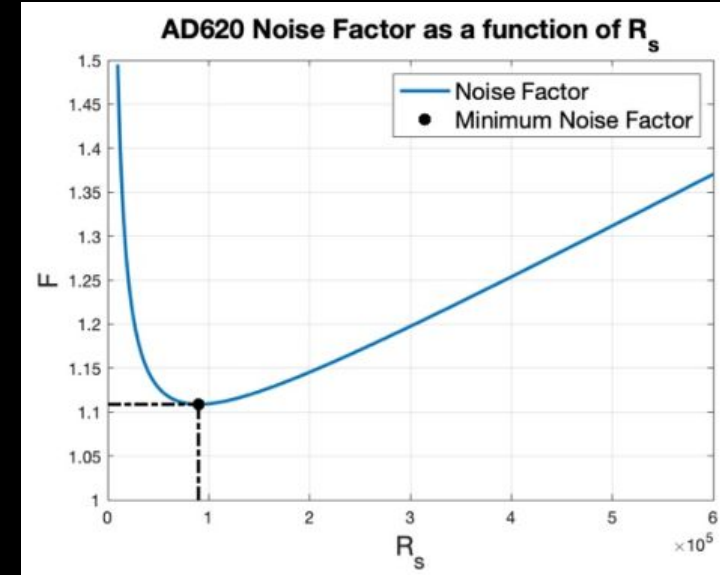
LM324 Source impedance = 4 MOhms

Thermal noise of highest resistor (200 kOhms) = 3.1 μ V

NF (AD620) = ~ 1.1 , SNR loss of 0.45 dB

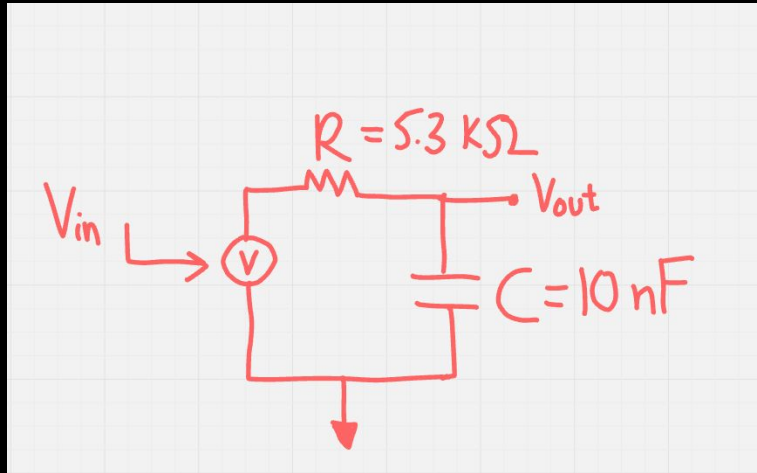
NF (LM324) = ~ 48 , SNR loss of 16.81 dB

60 Hz EMI $\rightarrow$ Notch filter to address dominating power line interference



Anti-Aliasing Filters and ADC

- Arduino acquired data saturated by amplified noise, so unable to see ABR
- Must have maintain a sampling rate of around 6 kHz
- Hard to perform averaging with low memory capacity
- Quantization error (insufficient vertical resolution): Smallest voltage change = (Full scale range) / $2^B = 5 / 2^{10} = 4.88 \text{ mV}$



$$F_c = 3 \text{ kHz}$$

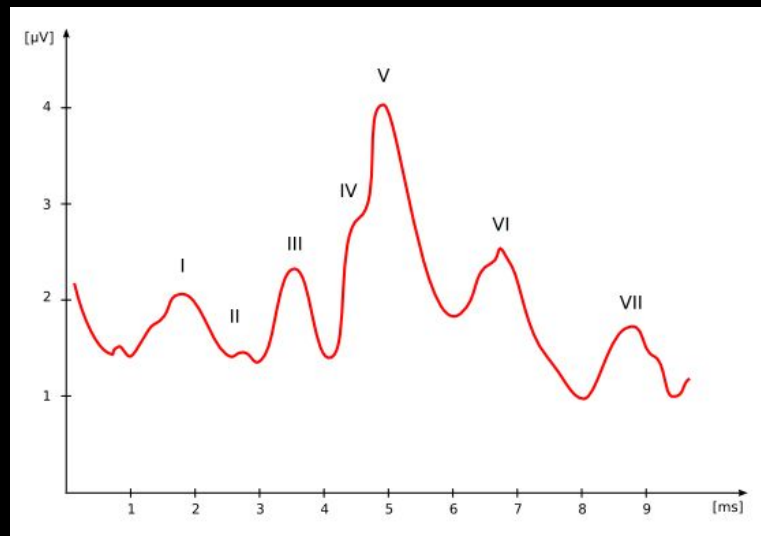
$V_{in} \rightarrow$ Output of circuit

$V_{out} \rightarrow$ Input to
arduino

ABR Comparison to Gold Standard



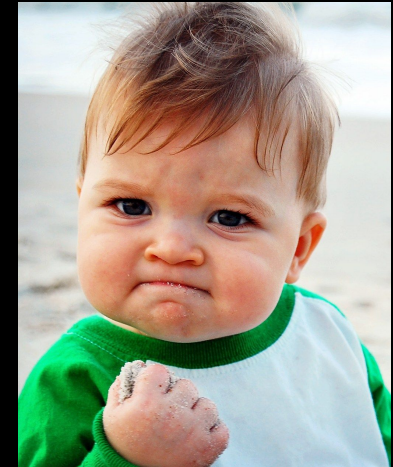
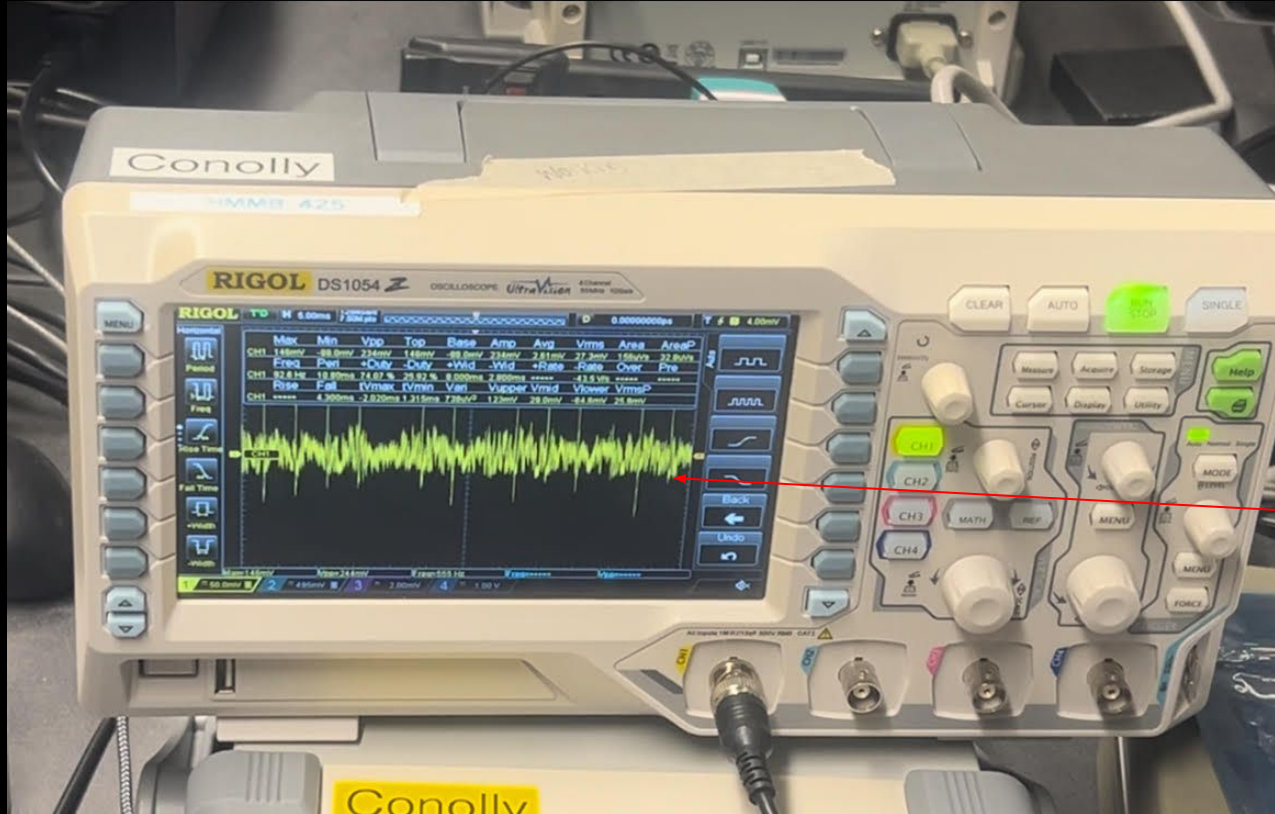
ABR Oscilloscope Waveform



Ideal ABR Waveform

SNR estimation: $V_{rms} / \text{noise std} = 20 * \log(7.071/0.3) = 27.5 \text{ dB}$
This means there are a LOT of artifacts

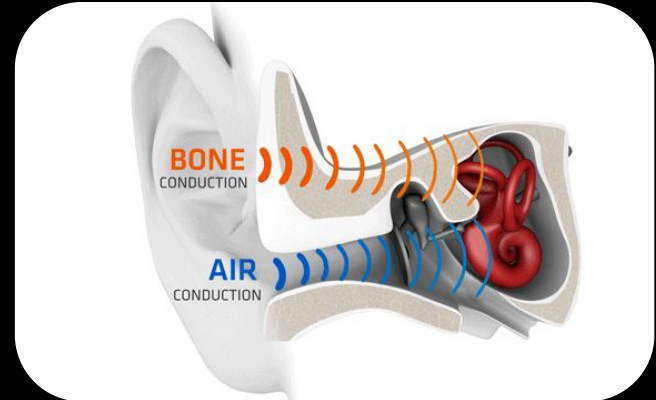
20 Hz - 20 kHz Electrode Vibration Test



Very high frequency as tone approaches 20 kHz

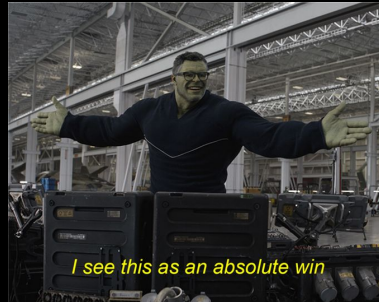
Tradeoffs and Lessons Learned

- Dry vs wet electrodes: dry still needs gel, wet works just as well
- LM324: high noise compounds at every gain stage
- ABR amplitude so small, skull vibrations large
- The amplitude of ABR is SMALL



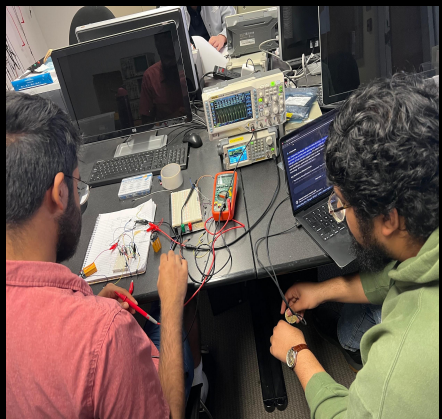
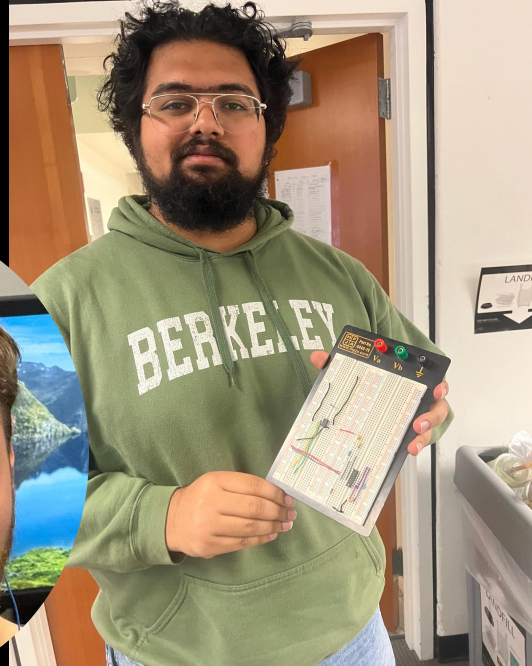
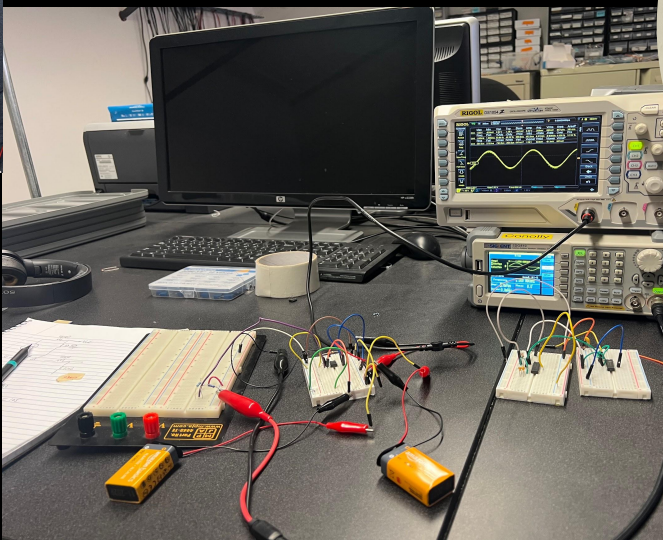
Conclusion:

- ABR: gold standard for catching infant hearing loss early
- Clinical devices cost \$8,000-\$20,000 for a reason
- Learned firsthand course concepts in practice
- Our signal responds to auditory stimulus, validating the core concept
- Same fundamentals as EKG, just a much smaller signal 😊





THANK YOU!



References:

- Analog Devices, "Low Cost Low Power Instrumentation Amplifier AD620," Rev. H, Jul. 2011.
- Texas Instruments, "LMx24-N, LM2902-N Low-Power, Quad-Operational Amplifiers," SNOSC16D, Jan. 2015.
- "Trial-By-Trial Auditory Brainstem Response Detection." *bioRxiv*, February 3, 2026. <https://doi.org/10.64898/2026.01.31.703019>
- Cost-effectiveness of portable-automated ABR for universal neonatal hearing screening in India. *Frontiers in Public Health*, 12. <https://doi.org/10.3389/fpubh.2024.1364226>
- Ma, J., Choi, S. J., Kim, S., & Hong, M. (2024). Performance comparison of convolutional neural network-based hearing loss classification model using auditory brainstem response data. *Diagnostics*, 14(12), Article 1232. <https://doi.org/10.3390/diagnostics14121232>